

Name: _____

Date: _____

SEQUENCES, (QUADRATIC SEQUENCES)

LO: I can find the nth term of a quadratic sequence.

What is a quadratic sequence?

A **quadratic sequence** has a constant **second difference**. The nth term has the form $an^2 + bn + c$.

Strategy: Find first differences, then second differences. If the second difference is constant ($= 2a$), the sequence is quadratic. Then find a, b, c.

Quick examples

●	1, 4, 9, 16, 25, ... has nth term n^2	second diff = 2
●	3, 8, 15, 24, 35, ... has 2nd diff = 2	quadratic
●	Second difference = $2a$, so if 2nd diff = 6 then $a = 3$	finding a
●	2, 5, 8, 11, ... is quadratic	constant first diff = arithmetic

Key idea

Find second differences ($= 2a$). Subtract an^2 from each term to get a linear sequence, then find b and c.

$$u_n = a n^2 + b n + c$$

$2a$ = second difference; then find b and c from the remainder

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - finding the nth term

Find the nth term of 3, 8, 15, 24, 35, ...

First differences: 5, 7, 9, 11. Second differences: 2, 2, 2. So $2a = 2$, $a = 1$

Subtract n^2 : $3-1=2$, $8-4=4$, $15-9=6$, $24-16=8$. Linear sequence: 2, 4, 6, 8
 $= 2n$

nth term = $n^2 + 2n$

$$n^2 + 2n$$

Check: when $n=1$: $1+2=3$. When $n=2$: $4+4=8$. Correct!

Example 2 - second difference not 2

Find the nth term of 2, 8, 18, 32, 50, ...

First diff: 6, 10, 14, 18. Second diff: 4, 4, 4. So $2a = 4$, $a = 2$

Subtract $2n^2$: $2-2=0$, $8-8=0$, $18-18=0$.

All zeros!

nth term = $2n^2$

$$2n^2$$

When subtracting an^2 gives all zeros, $b = c = 0$.

Example 3 - with a constant

Find the nth term of 4, 7, 12, 19, 28, ...

First diff: 3, 5, 7, 9. Second diff: 2, 2, 2. $a = 1$

Subtract n^2 : $4-1=3$, $7-4=3$, $12-9=3$, $19-16=3$. Constant = 3
nth term = $n^2 + 3$

$$n^2 + 3$$

The remainder is a constant, so $b = 0$ and $c = 3$.

Example 4 - full quadratic

Find the nth term of 2, 9, 20, 35, 54, ...

First diff: 7, 11, 15, 19. Second diff: 4, 4, 4. $a = 2$

Subtract $2n^2$: $2-2=0$, $9-8=1$, $20-18=2$, $35-32=3$. Linear: 0, 1, 2, 3 = $n - 1$
nth term = $2n^2 + n - 1$

$$2n^2 + n - 1$$

Check: $n=1$: $2+1-1=2$. $n=2$: $8+2-1=9$. Correct!

YOUR TURN – PRACTICE (deliberate practice)

1. Recognising quadratic sequences

Show these are quadratic by finding second differences.

- a) 1, 4, 9, 16, 25
- b) 2, 6, 12, 20, 30
- c) 3, 12, 27, 48, 75
- d) 0, 3, 10, 21, 36

2. Simple nth terms

Find the nth term.

- a) 1, 4, 9, 16, 25, ...
- b) 4, 7, 12, 19, 28, ...
- c) 0, 3, 8, 15, 24, ...
- d) 2, 8, 18, 32, 50, ...

3. With linear part

Find the nth term in the form $an^2 + bn + c$.

- a) 3, 8, 15, 24, 35, ...
- b) 2, 7, 14, 23, 34, ...
- c) 4, 10, 18, 28, 40, ...
- d) 5, 12, 23, 38, 57, ...

4. Harder nth terms

Find the nth term.

- a) 6, 15, 28, 45, 66, ...
- b) 1, 7, 17, 31, 49, ...
- c) 0, 6, 16, 30, 48, ...
- d) 5, 14, 27, 44, 65, ...

5. Mixed problems

Find the nth term and use it.

- a) 3, 6, 11, 18, 27, ... nth term?
- b) Using your answer to (a), find the 10th term.
- c) 2, 9, 20, 35, 54, ... nth term?
- d) Is 100 in the sequence $n^2 + 2n$?
- e) Find the 20th term of $3n^2 - n + 1$.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) A quadratic sequence has a constant second difference ☐

Correct answer: _____

Reason: _____

b) If first differences are constant, the sequence is quadratic ☐

Correct answer: _____

Reason: _____

c) Second difference = $2a$ where a is the coefficient of n^2 ☐

Correct answer: _____

Reason: _____

d) The second difference equals the coefficient of n^2 ☐

Correct answer: _____

Reason: _____

e) After subtracting an^2 , the remainder forms a linear sequence ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) The nth term of a quadratic sequence is $an^2 + bn + c$. The 1st term is 5, the 2nd term is 13, and the 3rd term is 25. Find a , b , and c .

Expression: _____

Simplified: _____

b) Prove that the difference between the nth and (n+1)th terms of $3n^2 + 2n + 1$ is always an odd number.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

